

AI-Powered EV Battery Fire Prevention System

Ensuring a Fire-Free, Secure & Sustainable EV Future

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EV.Engineer



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EV.Engineer, iTelematics Software Private Limited

Information Technology, NIT Karnataka, Surathkal

Electric Vehicle Engineering & Development, CODE, IIT Madras

AI-Powered EV Battery Fire Prevention System

The Problem

EV Battery Fires are a Major Concern

- Frequent thermal runaway incidents leading to fire hazards.
- Lack of real-time battery health monitoring & risk alerts.
- Fleet operators & EV owners suffer from unexpected breakdowns and expensive battery replacements.
- Regulatory pressure (AIS-156) for stricter safety measures.

Example : Bangalore has seen a 300% increase in EV fire incidents in 2023-24.

Project / Module Details

AI-Powered EV Battery Fire Prevention System

- Battery Temperature Monitoring System
- Battery Voltage & Current Analysis
- State of Charge (SOC) Estimation
- EV Battery Health Prediction
- Real-Time Battery Monitoring with IoT
- Intrusion Detection in Battery Management System (BMS)

The Solution

Leverages AI & Machine Learning to predict battery failures before thermal runaway.

Real-Time Monitoring of critical parameters:

- Temperature fluctuations
- Voltage imbalances
- Cell inconsistencies

AI-Driven Predictive Analytics for early detection of anomalies.

Automated Preventive Actions:

- Controlled discharge to prevent overheating
- Active cooling mechanisms (liquid/air cooling)
- Emergency shutdown & alerts

Seamless BMS Integration:

- Works with existing Battery Management Systems
- Adds AI-powered safety layer

Cloud-Based Analytics & OTA Updates:

- Continuous learning from real-world battery failures
- Over-the-Air (OTA) updates for AI model improvements

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Access devices / sensors from connected EV / Software Defined Vehicles

CONNECT	Connect to the Vehicle from Mobile device and Authenticate.
DETECT	Detect Devices & Sensors (Battery Details, Telematics Information.. etc)
READ	Read the status of the Devices & Sensors
WRITE	Change the device / sensor status
DISPLAY	Display Device / Sensor’s info on Dashboard
COLLECT	Collect and upload device details to Cloud for Analysis
ANALYSE	Device analysis using Machine Learning
CONTROL	Control vehicle using mobile (Lock, Unlock, Start, Stop | CAN Bus)

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Intrusion detection in connected EV / Software Defined Vehicles

SCAN	Scan the Devices | Sensors | Battery | Telematics | WiFi in the Vehicle (On demand basis)
MONITOR	Monitor the vehicle system for accidental attack
DETECT	Detect Intrusion of attack from Network | Internet | Other IoT | Apps
ALERT	Alert the user about the issues / problems
COLLECT	Collect and upload Intrusion details to Cloud for Analysis
ANALYSE	Intrusion analysis using Machine Learning
REPORT	Generate the report (Detected Issues and other analysis information)
RECOMMEND	Recommendation | Recovery | Protection

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Intrusion detection in Battery Management System

- Collect Battery Data Logs (or Use Sample Data)
- Analyse Normal vs. Anomalous Data
- Implement an Anomaly Detection Model
- Real-Time Intrusion Detection Simulation
- Secure Battery Data with Encryption

Potential Cyber Threats:

- **Spoofing Attack**: Fake voltage readings injected
- **Man-in-the-Middle Attack:** SOC data modified
- **Malware in BMS:** Unauthorised data manipulation

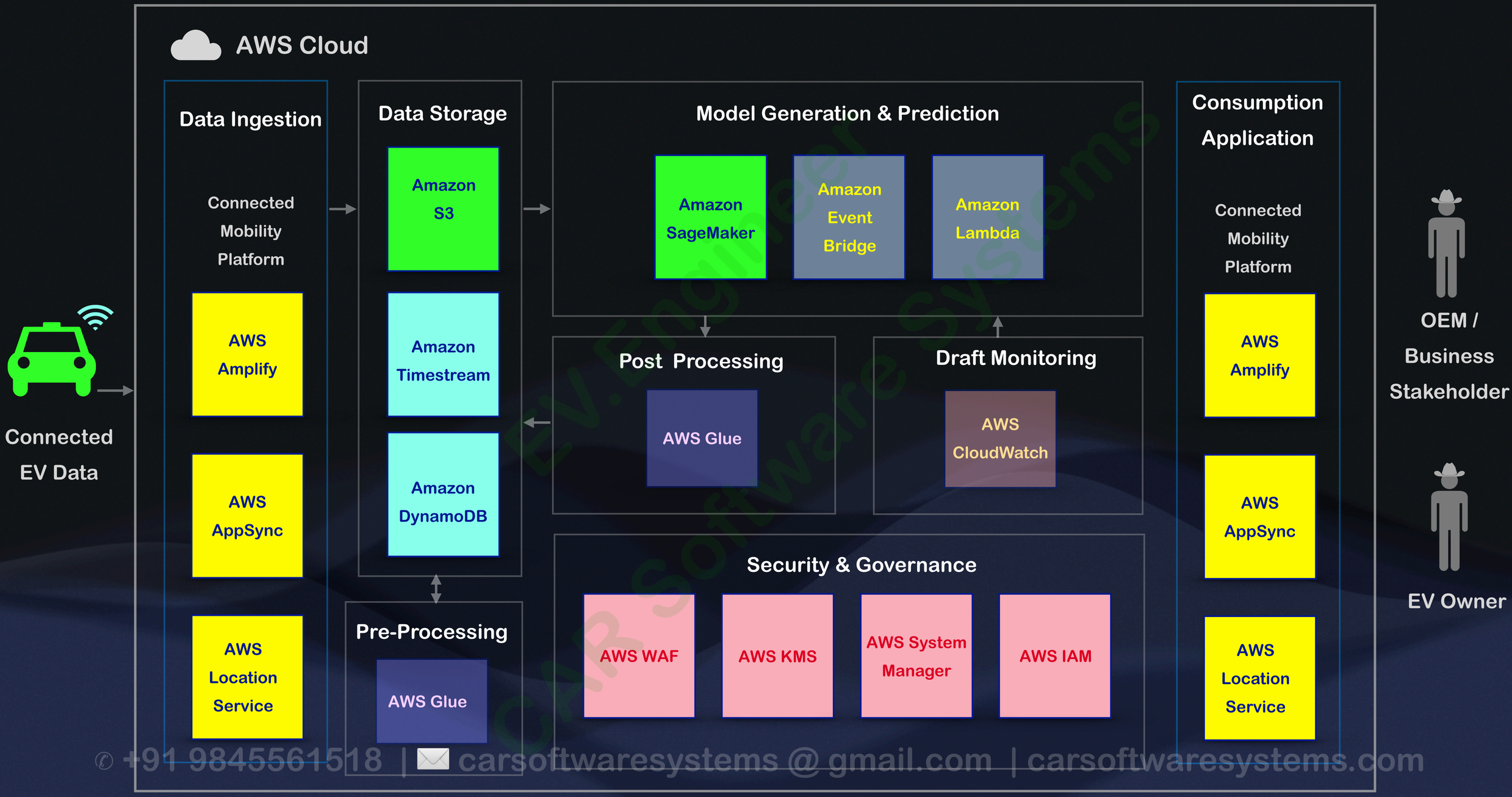
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Battery Diagnostics Reports / Fault Status

Short Circuit	Deep Discharge	Health
Imbalance	Over Heating	Safety Level
BMS Fault	Reserve Current	Energy Level
Charger Fault	Voltage Drop	SOC
Aging	Water Damage	SOH

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Cloud Architectural Design



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Architecture of EV Battery Management Systems

INPUT

Battery Pack Parameters
- Current
- Voltage
- Temperature

Motor Inputs
- Power
- Torque demand

Surrounding Input
- Temperature
- Humidity

Telematics
- Data from Telematics Control Units

Other Inputs
- Power demand from Auxiliary units

SENSOR INTERFACE

- Current Sensor
- Voltage Sensor
- Temperature Sensor

BATTERY MODEL

- Electrochemical / EMC
- Thermal Model

SAFETY MANAGEMENT SYSTEM

- On-board diagnostics
- Safety Controls

Charging controller

THERMAL MANAGEMENT

Execution of cooling method
- Passive
- Active
- Adaptive

CELL BALANCING MODULE

Execution of Balancing topology
- Passive / Dissipative
- Active / non-dissipative

Memory Storage

STATE ESTIMATION MODULE

- SOT Estimation
- SOP Estimation
- SOC Estimation
- SOH Estimation
- RuL Prediction
  - DOD
  - EOD

DOD (Degradation Onset Detection)
EOD (End of Degradation)

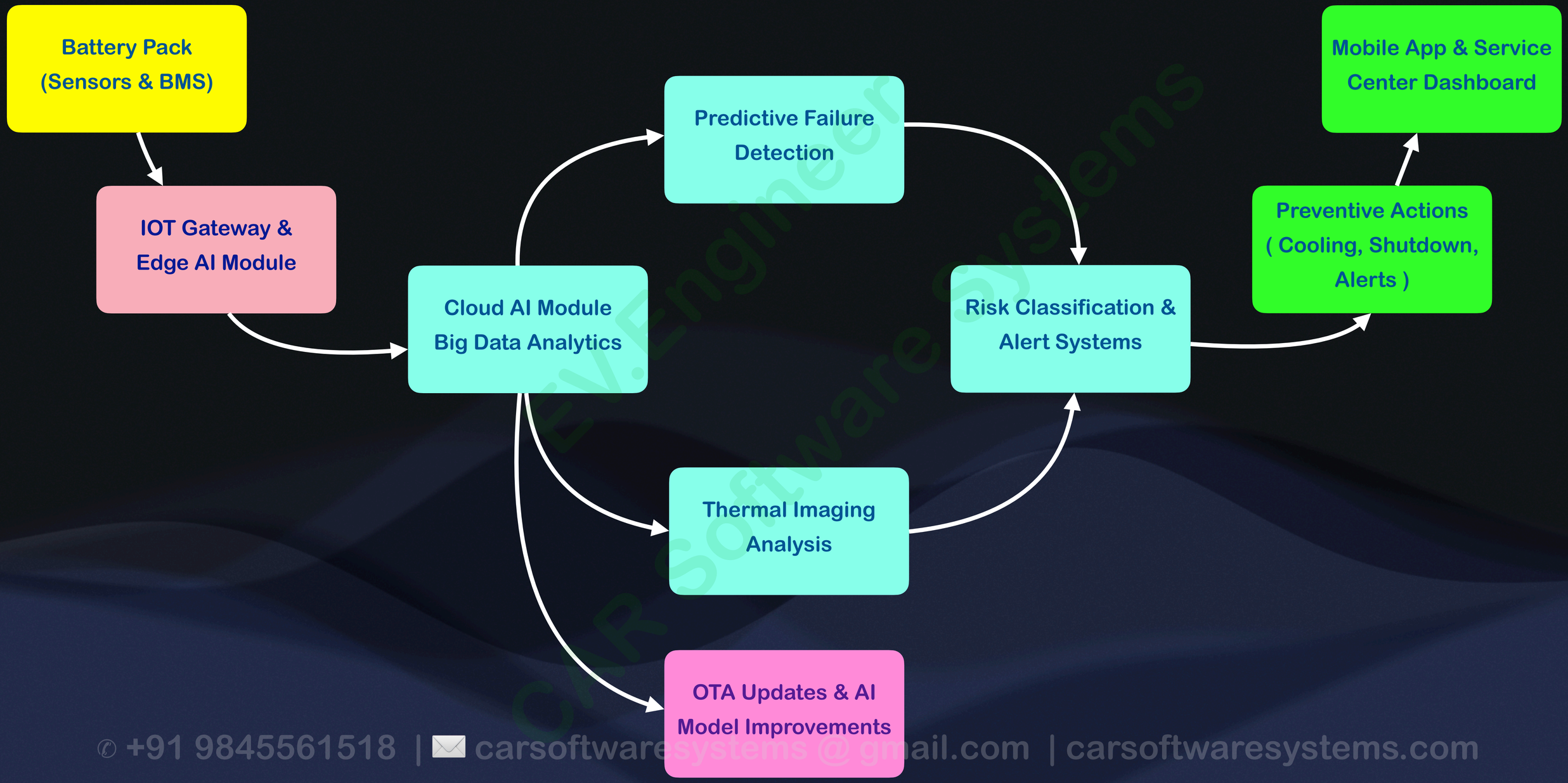
CAN BUS

OUTPUT

- Vehicle Control Unit
- Motor Control Unit
- Cell Balancing Circuit
- Dashboard
- Diagnostic Port
- Safely Functions
- Cooling System

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AI & Quantum Computing - Layered Architecture

Cloud & Edge Quantum Computing Infrastructure

(Top-Level Control & Computation)

Secure Quantum Cryptography Layer

(Ensures Data Integrity & Security)

AI-Powered Anomaly Detection & Prediction Layer

(Early Warning System - Classical AI Approach)

Quantum Computing Optimisation & Decision Making

(Advanced AI with Quantum Computing)

EV Battery Data Collection & Monitoring Layer

(Real-Time Execution & Sensor Data Processing)

Hybrid Quantum - Classical AI System :

- Uses IBM Qiskit, Microsoft Azure Quantum, Google Cirq for cloud-based quantum simulations.
- Supports real-time Quantum AI execution for battery analytics.
- Balances computational workload between Classical AI and Quantum AI for optimised processing.

Quantum Edge Computing for Real-Time Battery Monitoring

- Processes data locally for fast response and battery failure prevention.
- Reduces latency by executing Quantum AI models at the edge.

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Quantum Key Distribution (QKD) for Secure Over-the-Air (OTA) Updates

- Ensures BMS firmware updates remain protected against cyber threats.
- Integrates with AI-driven cybersecurity to detect and mitigate potential breaches.

Post-Quantum Cryptography (PQC) for Secure EV Data Storage

- Encrypts battery logs, BMS firmware, and user data to prevent hacking.
- Provides resilience against classical and quantum cyber threats.

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Classical AI/ML for Initial Anomaly Detection

- AI models trained on historical EV battery failure incidents.
- Identifies early warning signs of thermal runaway.
- Uses probabilistic models and deep learning for failure prediction.

Deep Learning for Fire Risk Estimation

- Neural Networks classify battery safety levels and generate alerts.
- Implements explainable AI (XAI) to interpret failure causes.

Classical Optimisation Algorithms for Battery Management

- Uses Reinforcement Learning & Heuristic Search to optimize battery efficiency.
- Enhances battery longevity and optimal energy usage.

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Quantum Machine Learning (QML) for Battery Health Prediction

- Uses Variational Quantum Circuits (VQC) for complex pattern recognition.
- Enhances AI's ability to process non-linear battery degradation patterns.

Quantum Neural Networks (QNNs) for Thermal Runaway Risk Assessment

- Quantum-enhanced deep learning models predict potential failures.
- Simulates high-dimensional battery behaviour for precise anomaly detection.

Quantum Approximate Optimisation Algorithm (QAOA) for Energy Management

- Optimises battery charging, discharging, and thermal management.
- Uses quantum annealing techniques for highly efficient decision-making.

Quantum Annealing for Battery Safety Optimisation

- Uses D-Wave's quantum annealers for efficient battery performance tuning.
- Applies quantum-enhanced combinatorial optimisation for fire prevention strategies.

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(Real-Time Execution & Sensor Data Processing)

Real-time Sensor Data Acquisition

- Captures data from EV battery sensors (temperature, voltage, current, SOC, SOH).
- Uses IoT & Edge Computing at the Battery Management System (BMS) for real-time processing.
- Implements self-healing AI models that adapt to sensor noise and environmental variations.

Edge Computing at BMS

- Low-latency, real-time analysis to detect early battery anomalies.
- Integrates AI-driven edge computing for preemptive failure response.

Secure Data Transmission:

- Utilises Quantum Cryptography (QKD) for secure communication between EV and cloud servers.
- Ensures tamper-proof data logging for compliance and traceability.

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1. Battery Temperature Monitoring System

Goal: Read temperature data, analyse trends, and detect overheating.

Concepts: File handling, NumPy, Pandas, Matplotlib

Tasks:

- Read a CSV file containing battery temperature data
- Calculate average, max, and min temperatures
- Plot a temperature trend graph using Matplotlib
- Detect overheating conditions (e.g., alert if temp > 60°C)

Outcome: Basic battery monitoring using Python

2. Battery Voltage & Current Analysis

Goal: Analyse voltage & current data to detect anomalies..

Concepts: Pandas, Data Visualisation, Time-Series Analysis

Tasks:

- Load battery voltage & current datasets
- Identify voltage drops and current spikes
- Plot Voltage vs. Time & Current vs. Time
- Set a rule: Alert if voltage drops below a threshold

Outcome: Detect battery performance issues

3. State of Charge (SOC) Estimation

Goal: Estimate battery SOC using voltage and current data.

Concepts: Numerical computing, Basic Machine Learning

Tasks:

- Load historical battery data (Voltage, Current, SOC)
- Train a simple regression model to predict SOC
- Validate results using test data
- Display real-time SOC values for a given input

Outcome: SOC estimation using Python

4. EV Battery Health Prediction

Goal: Use AI to predict battery degradation over time.

Concepts: Machine Learning, Data Science

Tasks:

- Load battery charge-discharge cycle data
- Identify patterns in battery degradation
- Train an ML model (Scikit-learn) to predict Remaining Useful Life
- Visualise predictions with graphs

Outcome: AI-based battery health prediction

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5. Intrusion Detection in Battery Management System

Goal : Detect anomalous activities in an EV Battery Management System using Python. (Hacking attempts, data tampering, or unauthorised access)

Concepts Used:

- Log Analysis & Data Forensics
- Anomaly Detection (Machine Learning)
- Cybersecurity Threat Detection

Project Overview

The Battery Management System (BMS) logs critical parameters:

- Voltage, Current, Temperature
- State of Charge (SOC), State of Health (SOH)
- Communication logs (CAN messages)

Potential Cyber Threats:

- **Spoofing Attack**: Fake voltage readings injected
- **Man-in-the-Middle Attack:** SOC data modified
- **Malware in BMS:** Unauthorised data manipulation

Expected Outcomes

- Build a Battery Intrusion Detection System (IDS)
- Detect cyber attacks on BMS data
- Train an ML model to differentiate between normal and attack conditions
- Secure BMS communication with encryption (Advanced)

STEP 1 : Collect Battery Data Logs (or Use Sample Data)

- Use a CSV file containing battery logs with timestamps
- Add a column for intrusion detection labels (Normal / Attack)

STEP 2 : Analyse Normal vs. Anomalous Data

- Load the dataset using Pandas
- Visualise voltage/current variations using Matplotlib
- Identify unexpected spikes, drops, or inconsistent SOC values

STEP 3 : Implement an Anomaly Detection Model

- Use Scikit-Learn to train an ML model for intrusion detection
- Algorithms: Isolation Forest, Random Forest, or Logistic Regression
- Train model on normal vs. attack data samples
- Detect real-time anomalies from live battery logs

STEP 4 : Real-Time Intrusion Detection Simulation

- Simulate incoming battery data (live stream using Python)
- Detect unauthorised activities and trigger alerts
- Implement logging system to save security breach attempts

STEP 5 : Secure Battery Data with Encryption

- Use AES Encryption (Python pycryptodome module)
- Encrypt critical BMS data before transmission \
- Ensure only authorised systems can decrypt it

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Business Plan



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- Frequent **thermal runaway incidents** leading to fire hazards.
- Lack of **real-time battery health monitoring** & **risk alerts.**
- **Fleet operators & EV owners** suffer from unexpected breakdowns and expensive battery replacements.
- **Regulatory pressure** (AIS-156) for stricter safety measures.

Example : Bangalore has seen a **300% increase in EV fire incidents in 2023-24.**

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The Solution

AI - Powered Thermal Runaway Early Warning System (TREWS)

- **Predicts & prevents battery overheating & fire risks** using real-time analytics.
- **AI-driven thermal modelling** detects early failure patterns.
- **Smart charging optimisation** prevents excessive heat buildup.
- **Instant alerts via SMS, WhatsApp, Fleet Dashboards** for preemptive action.
- **Cybersecurity integration** to prevent data tampering & enhance safety.

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Vision Statement

"To revolutionise EV Battery safety with AI-powered predictive technology, ensuring a fire-free, secure, and sustainable electric mobility future."

Mission Statement

"We are building AI-driven early warning systems that predict and prevent EV Battery Thermal runaway, ensuring a fire-free and secure electric mobility future.

Our solution reduces risks, enhances battery longevity, and provides real-time safety insights for fleet operators, service centers, and manufacturers—creating a scalable, high-impact business model in the growing EV industry."

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Core Values

- **Innovation** – We push boundaries in AI and predictive analytics to enhance EV safety.
- **Quality** – Delivering accurate, reliable, and high-performance safety solutions.
- **Safety & Security** – Protecting lives and assets by preventing battery failures before they happen.
- **Customer-Centricity** – Focused on solving real-world EV battery safety challenges for fleet operators, service centers, and manufacturers.
- **Team Empowerment** – Fostering a culture of excellence, collaboration, and continuous learning.
- **Sustainability** – Promoting a cleaner, greener future by improving EV battery efficiency and longevity.

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Market Opportunity

EV Market Growth in India (2024-2030):

- 4M+ EVs on Indian roads by 2025 → Rising demand for predictive safety solutions.
- $2B+ TAM in EV battery analytics & predictive safety by 2027
- EV Two-Wheeler Market CAGR 49% → 90% of growth from fleet & delivery startups.
- Battery Fire Incidents Increased by 300% (2023-24) → Regulatory compliance & safety demands.
- Total Addressable Market : $2B+ EV Battery Analytics Industry by 2027.

Primary Target

- Bangalore's EV Battery Service Centers & Fleets → Rapid Expansion to Other Cities.

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Target Customers & Business Model

Target Customers

- **EV Battery Service Centres** – Main focus for early adoption & pilot testing.
- **Fleet Operators** – Require predictive battery health insights.
- **Battery Swapping Networks** – Need real-time monitoring for multiple batteries.
- **EV OEMs & Dealerships** – Long-term partnerships for factory-level integration.

Business Model

- **Service Center Subscription** : ₹2,999 - ₹9,999 per month (for battery analytics & fire prevention insights)
- **Add-on Services for EV Owners** (via Service Centres) : Freemium model (basic free, premium ₹499/year)
- **API Licensing for EV OEMs & Battery Swapping Companies.**

Goal

- **Secure 3-5 pilot customers in Bangalore → Scale Nationwide.**

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Competitive Landscape

Competitor	Solution	Weakness
Ola Electric	Internal battery monitoring	No AI-based thermal runaway prediction
Ather Energy	BMS safety system	No external predictive analytics
ION Energy	AI analytics for OEMs	Not available for individual EV owners
Log9 Materials	Battery R&D & safety focus	No real-time user alerts
EV Doctor	AI-powered battery diagnostics & monitoring	Primarily targets service centres; limited end-user focus
Our Solution	AI-powered real-time battery fire prevention system	First with proactive early warning & smart alert system
	Key Differentiators : Predictive AI | Smart Alerts | Cybersecurity | Scalable SaaS.	

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Roadmap & Execution Plan

Short-Term (0-1 Year):

- Secure partnerships with 3+ major battery service centers in Bangalore for pilot testing.
- Deploy AI-based predictive analytics as a service for EV battery inspections.
- Collect real-world data & refine AI model accuracy.

Medium-Term (1-3 Years):

- Scale to 50+ service centers across major Indian cities.
- Expand to battery swapping companies & fleet operators.
- Integrate with OEMs for factory-level safety compliance.

Long-Term (3-5 Years):

- Become India's #1 AI-driven EV battery safety platform.
- Expand beyond 2-wheelers to 4-wheelers & public transport.
- Standardise AI-driven battery diagnostics across all EV service centers

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Risk Management

- **Dependence on Data Availability** – AI accuracy depends on access to high-quality battery telemetry data from service centers & fleets.
- **Service Center Adoption Barrier** – Some traditional service centers may resist adopting AI-based predictive solutions due to a lack of familiarity.
- **Integration with Existing BMS Systems** – Many EVs have built-in BMS; we need to demonstrate the added value of our AI analytics.
- **Cybersecurity Risks** – Handling critical battery safety data makes the system a potential target for hacking or tampering.
- **Real-Time Processing Costs** – Running AI-driven analytics at scale requires cloud infrastructure, balancing costs while keeping subscriptions affordable.

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Mitigation Strategies

- Partner with leading fleet operators & service centers to improve data collection.
- Provide training & easy-to-use interfaces for service centers.
- Highlight differentiation from BMS by offering early risk detection & alerts.
- Implement strong encryption & cybersecurity protocols to protect data.
- Optimise cloud-based architecture to balance cost efficiency & performance.

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Investment & Funding Requirements

- Develop & test AI models with real-world data
- Build the web platform & API integrations.
- Launch pilot programs with fleet operators & service centers.
- Scale cybersecurity & cloud infrastructure.

Potential Investors & Grants

- Micelio Fund, Blume Ventures, Indian Angel Network.
- **Government Grants**: FAME India, Startup India, NITI Aayog.
- **Corporate Collaborations**: EV battery makers, fleet operators, OEMs.

Goal

- Secure first funding → Build MVP → Achieve Product-Market Fit → Scale Nationwide.

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Join Us in Creating a Fire-Free EV Future!

Looking for Strategic Partners, Pilot Customers & Investors.

Thank you

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